KubeEdgeInfer

A Closed-Loop, eBPF-Driven Kubernetes Framework for Heterogeneous Distributed LLM Inference on Consumer Edge Clusters

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Introduction

- Large Language Models (LLMs) are often too large to fit on a single consumer GPU.
- **Pipeline parallelism** solves this by splitting model layers across several machines, forming an inference assembly line.
- Existing frameworks are built either for stable datacenter interconnects, or they split layers **statically** and never revisit that decision.
- Consumer and edge hardware is inherently **heterogeneous**: different GPU generations, WiFi rather than Ethernet, and thermal variance under sustained load.

Core Idea

Continuously observe real hardware conditions at the kernel level, and let the model's layer partition *adapt* to them instead of staying fixed.

Literature Review

Work	Year	Contribution	Gap this work addresses
EdgeShard	2024	DP device selection and layer partition	Profiled offline; plan never revisited
Galaxy	2024	Tensor/sequence parallelism with overlap	One-shot planning, no adaptation
Helix	2025	Max-flow MILP over mixed GPUs	Solver too heavy for a live loop
TPI-LLM	2024	Memory scheduling for 70B on small RAM	Optimises memory, not degradation
prima.cpp	2025	CPU/GPU co-scheduling on home clusters	Schedule fixed at load time
Head-level split	2025	Attention-head granularity, with migration	Triggered by memory pressure only
Parallax	2025	Serving over volunteer machines	No kernel-level measurement
Adaptive split (MBRL)	2025	Learns the split point as the channel varies	Needs training; no optimality proof
Distributed LLM survey	2025	Surveys the field	Names adaptation as an open problem
sched_ext (eBPF)	2025	Linux schedulers loaded as eBPF programs	Single host; no model pipeline

Common limitation

Every one of these measures the hardware *before* the run and never measures again.

Problem Statement

Static partitioning ignores real-time hardware variability.

Thermal Throttling

GPUs slow down mid-inference as they heat up under sustained load.

Network Latency

WiFi and edge links have variable latency — no InfiniBand or NVLink at home.

Node Failure

A laptop closes its lid or drops off the network, and its layers go with it.

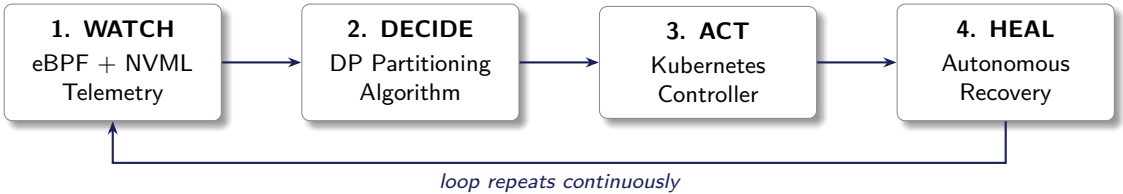
Consequence: Pipeline Bubble Time

Fast machines sit idle waiting on slower ones. A one-time, fixed split cannot correct itself as conditions change.

Objectives and Contributions

1. Capture system telemetry — network transfer latency, GPU thermal state, throttling — using **eBPF** and **NVML**, with no change to the model code.
2. Design a **dynamic partitioning algorithm** that computes the optimal contiguous layer split from live telemetry, and prove it optimal for each snapshot.
3. Integrate telemetry and partitioning decisions directly into a **Kubernetes** control loop, applying them without restarting pods.
4. Enable **autonomous healing**: automatic re-partitioning on node failure or sustained thermal throttling.
5. Stabilise the loop with **hysteresis** — a 15% improvement threshold and a 30 s cooldown — so noisy telemetry cannot cause thrashing.
6. Evaluate bubble time, throughput and time-to-first-token empirically against static partitioning and an offline-profiling baseline.

Architecture — How It Works



- **Watch:** kernel-level flow latency, GPU temperature, memory, throttle state.
- **Decide:** compute the optimal contiguous split from live metrics.
- **Act:** re-assign layers, reroute traffic, apply the new split with no restart.
- **Heal:** detect node failure within 3 s, re-partition across the survivors.

Results

Injected condition	KubeEdgeInfer	Profile-once	Static split
No fault	42 ms	42 ms	42 ms
Thermal throttle (0.4×)	**52 ms**	102 ms	102 ms
Network delay (80 ms)	**62 ms**	122 ms	122 ms
Thermal + network	**62 ms**	182 ms	182 ms
Node failure	**62 ms**	62 ms	pipeline stops

Bottleneck-stage cost per token; lower is better. 12 layers over 3 machines.

Headline

49% lower bottleneck cost under thermal throttling and under network degradation; **66%** lower when both occur together. Under node failure the static split is undefined over the surviving set, so the pipeline stops — KubeEdgeInfer re-partitions and continues.

Reproducible with `go run ./bench/dpsim`; end-to-end throughput comes from `bench/run.py`, which needs a live cluster.

Novelty

Feature	EdgeShard	Galaxy	Helix	prima.cpp	Ours
Heterogeneous edge support	✓	✓	✗	✓	✓
Kernel-level telemetry (eBPF)	✗	✗	✗	✗	✓
Dynamic re-partitioning	✗	✗	✗	✗	✓
Thermal awareness	✗	✗	✗	✗	✓
Autonomous healing	✗	✗	✓	✗	✓
Declarative K8s API	✗	✗	✗	✗	✓

One-line novelty statement

We turn a static, one-time model-splitting decision into a continuous, self-correcting one — driven by real kernel-level measurements instead of assumptions.

Summary

- First system to close the loop: eBPF/NVML telemetry → Kubernetes scheduling → self-healing, for edge LLM inference.
- The partitioner is **provably optimal** for a fixed telemetry snapshot — it reduces to the classical linear partition problem, and is unit-tested against brute force on 200 random instances.
- Adaptive behaviour is validated **empirically** through a staged ablation under simulated network, thermal and failure conditions.
- Distributed GPT-2 output is **token-identical** to single-process inference, so adaptivity costs no correctness.
- Fully software-based and reproducible — no proprietary hardware lab required.

Thank you — Questions?

References

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Tools: cilium/ebpf (CO-RE loader), Kubernetes controller-runtime, NVIDIA NVML, k3s, Hugging Face Transformers.